

Michael Samaan Anwar

Data Scientist

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Summary

Data Scientist & ML Engineer experienced in end-to-end data pipelines—from exploratory analysis and data preprocessing to predictive modeling and deployment. Combines analytical rigor using SQL, Pandas, NumPy, and Power BI with advanced modeling in PyTorch and XGBoost to deliver data-driven insight and computer vision systems.

Education

Modern Academy, Maadi

Graduate
Cairo, Egypt

Experience

Machine Learning Engineer Trainee (Microsoft Track), DEPI – Digital Egypt Pioneers Initiative 07/2026 - Present

- Training on the Microsoft Machine Learning Engineer track of Egypt's national Digital Egypt Pioneers Initiative, an applied programme focused on building production-oriented machine learning skills

AI Training Program Participant, Sharks 07/2023 - 09/2023

- Explored foundational machine learning concepts and supervised classification models, implementing algorithms such as Support Vector Machines (SVM).
- Formulated data exploration and visualization pipelines, completing a capstone classification project on the Iris benchmark dataset to predict species characteristics.

AI Training Program Participant, IT Gate 07/2022 - 09/2022

- Completed an intensive 2-Months foundational training on core Python development and interactive environment setups using Jupyter Notebooks.
- Practiced foundational data analysis techniques using NumPy and Pandas for data manipulation, structured cleaning, and introductory statistical visualization.

Projects

Football Computer Vision System

Technologies: Python, OpenCV, YOLOv8, ByteTrack, PyTorch

- **Trained** a YOLOv8s object detection model across 3,052 images and 4,835 instances, achieving 84.8% mAP50 and 84.9% Precision.
- **Implemented** ByteTrack multi-object tracking to process real-time match analytics at 25 FPS.

Football Scout REST API

Technologies: Python, Flask, XGBoost, Scikit-Learn, Docker

- **Built** an ML-driven scouting platform analyzing 1,469 professional players, achieving 99.2% recommendation accuracy.
- **Deployed** an XGBoost valuation model ($R^2 = 0.753$, $MAE = €6.29M$) via a Flask REST API interface.

Expected Goals (xG) Shot Quality Pipeline

Technologies: Python, Scikit-Learn, Pandas

Designed and evaluated classification models estimating football shot quality and probability

NYC Airbnb Market Data Analysis

Technologies: Python, Pandas, NumPy, Power BI, Excel

- Cleaned and structured 40,000+ raw market listings, engineering custom location and pricing feature sets to analyze urban rental trends.
- Built interactive Power BI dashboards with slicers, KPI cards, and custom DAX measures to surface spatial pricing patterns and dynamic revenue drivers.

Skills

Programming: Python, C++, SQL, HTML/CSS

Machine Learning and AI: Scikit-Learn, PyTorch, TensorFlow, XGBoost, Random Forest, Feature Engineering

Computer Vision & Data Engineering: OpenCV, YOLO (YOLOv8), Object Detection, Multi-Object Tracking (ByteTrack)

Data and Analytics: Pandas, NumPy, Matplotlib, Seaborn, Power BI (Power Query, DAX)

Databases and Backend: MySQL, REST APIs, Flask

Cloud and Tools: GitHub, VS Code

Soft Skills: Technical Problem-Solving, Analytical Thinking, Data-Driven Strategy, Agile Collaboration, Adaptability

Certifications

- Supervised Machine Learning: Regression and Classification – Stanford / DeepLearning.AI [[Credential](#)]
- Python for Data Science & AI – IBM [[Credential](#)]
- Databases and SQL for Data Science with Python – IBM [[Credential](#)]
- Power BI Business Analytics – UC Davis

Languages

Arabic native ·

English